

Domain Model in Object-Oriented Analysis

Definition

A **Domain Model** is a visual representation of *conceptual classes* or real-situation objects in a domain. It illustrates noteworthy abstractions and relationships in the problem space without defining software methods or responsibilities. Also called *conceptual models*, *domain object models*, or *analysis object models*.

Purpose and Scope

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- Focuses on **things and products important to a business domain** (e.g., POS items, members, orders).
- Provides a **conceptual perspective** of a single domain rather than the entire business.
- Unlike a Business Object Model (BOM), a domain model avoids excessive up-front modeling.

Visual Representation

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- Uses UML **class diagrams** without methods.
- Illustrates:
 - Domain objects / conceptual classes
 - Associations between classes
 - Attributes of conceptual classes
- Serves as a **visual dictionary** to understand domain vocabulary and relationships.

Important Notes

Key Clarifications

- **Not a data model:** Does not show persistent storage or database structures.
- **Not software classes:** Does not include methods, responsibilities, or artifacts like windows or databases (unless modeling software domain concepts).
- **Conceptual classes:** Represent real-world concepts, ideas, or objects.

Conceptual Classes

Conceptual Classes

A conceptual class has three aspects:

1. **Symbol:** Name or representation (word or image), e.g., **Sale**.
2. **Intension:** Definition or description, e.g., “Represents the event of a purchase transaction, with date and time.”
3. **Extension:** Set of real-world instances, e.g., all sales occurring in the business.

Traditional Meanings of Domain Model

Traditional Meanings

- Conceptual perspective of real-world domain objects (focus of UP).
- Software perspective (commonly in Smalltalk / OO development): domain layer of software objects representing business logic.
- Both meanings are valid; context should clarify the intended meaning.

Summary

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- The domain model is a **visual dictionary** of the domain, helping analysts and developers understand the concepts, relationships, and terminology.
- It focuses on **real-situation conceptual classes**, not software artifacts or database structures.
- Conceptual classes may have attributes, purely behavioral roles, or even no attributes.